Review addendum: PR #1189

Date: 2026-10-29. Transfer quartic splitting from one side at five new primes.

Original author head: 93b20f3a3ca846f1d297432689d9512ec962b969. The original study and every frozen source, protocol, result, failed attempt and timing cap are preserved. This addendum takes precedence over unqualified experimental-status language in the archived report.

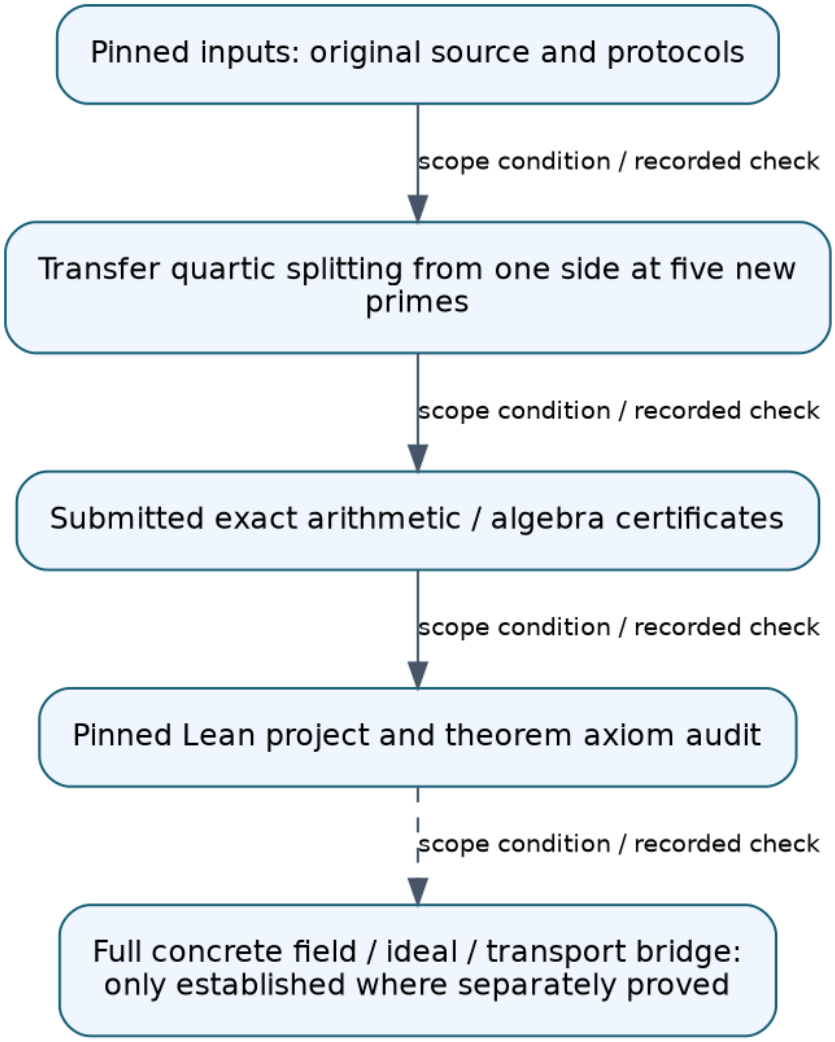
Recorded finding and its scope

The exact s -> -s conjugacy let one quartic supply **all 68** local splitting predictions for the other quartic at rational primes 41, 43, 47, 53, and 59. The prediction file was SHA-256 sealed before any right-quartic observation. All 68 predictions matched: 62 were checked by squarefree right-quartic residue factorization. At the six repeated reductions at primes 43 and 59, a normalized Taylor quadratic resolved the double-root cluster directly from the right quartic; right-side Kummer unit characters independently agreed. This gives a complete method-labelled local check across the five-prime set without factoring the right quartic in the prediction stage.

The paragraph above describes the retained certificate record, not a new experiment. Source: experiments/hecke-432-j1728-single-sided-splitting-20261026/RESULT.md (../../experiments/hecke-432-j1728-single-sided-splitting-20261026/RESULT.md). This review treats publication of protocol and results together as retrospective/exploratory unless an independent earlier publication witness is supplied. An archived PASS_PROSPECTIVE status is a literal historical label, not evidence of pre-observation registration. In particular, PR #1164's six-row holdout and PR #1190's 62/62 correctness checks retain their numerical results with this qualification.

Evidence-linked explanation

Evidence and proof boundaries for this study. Symbols C, B, E and K denote the frozen coordinate models or coefficient field, not newly registered subgroup instances. This diagram records algebra/certificate dependencies; it does not introduce a verified isogeny edge, EC1 identity, ISO1 route, or ECDLP transport claim.



Editable source: 1189.dot (1189.dot). Report and visual: 1189.pdf (1189.pdf).

Replay and proof map

The exact manifest binds 31 files in this study. The shared project pins Lean leanprover/lean4:v4.35.0-rc4 and Mathlib 021c46bdd042f67b538b9f9871d556a93713a9a9. All 37 original modules, 113 requested declaration reports and 142 noninternal theorems passed the integrated proof replay; this does not formalize every concrete Sage/PARI field or ideal interpretation. The source manifest and proof map are tools/ci/quartic_certificates.json (../../tools/ci/quartic_certificates.json) and the original integration report (../ci-repair-quartic-20261027.md).

This study's original modules: see the shared manifest; no new theorem is introduced by this addendum.

A new portable replay ran all 42 original integer auditors successfully in a disposable canonical-layout container. The checkout was read-only and all 1,404 original input hashes were verified before and after replay. See portable replay evidence (portable-replay/results.json). The separate field-index successor runner uses new output directories, unchanged source/protocol bytes, original timeout limits and the checked Sage launcher. Preparation and I/O controls are not a new Sage experiment.

Remaining obligations and graph review

Arithmetic table certificates and universal ring/permutation identities retain their precise hypotheses. Concrete number-field encodings, integral bases, ideals, Hensel lifting and Frobenius interpretation require

their own bridges. No security, speedup, prospective-validation or general impossibility claim is added.

The candidate catalog, volcano-IC figure set and paper figures were checked for affected links. This is a review and portability correction with no new surviving graphable mathematical fact; canonical graphs remain unchanged. The precise checked-path inventory is graph-review.json (graph-review.json).